

A Computer Model of Microplastic Damage to Neurons: How Plastic Build-Up May Slow Down Brain Signals

Abstract

Microplastics are tiny pieces of plastic that are now found almost everywhere, including inside the human body. Scientists do not yet know whether these particles harm the brain. This project uses a computer simulation to test whether plastic build-up inside a neuron's outer layer could slow down or weaken the electrical signals that brain cells use to communicate. The Hodgkin-Huxley model, a well-known mathematical description of how neurons fire, was adapted to include a variable for plastic build-up. Four scenarios were tested: a healthy neuron and three neurons with increasing levels of simulated plastic (1.5 times, 2.0 times, and 2.5 times the normal membrane capacitance). Each simulation ran for 60 milliseconds (ms) with a 10-microampere per square centimetre ($\mu\text{A}/\text{cm}^2$) electrical stimulus applied between 10 ms and 50 ms. The most contaminated neuron fired its first signal 17.18% later than the healthy neuron (14.19 ms compared to 12.11 ms) and reached a peak voltage 6.67 millivolts (mV) lower (34.06 mV compared to 40.73 mV). These results suggest that microplastic build-up could gradually weaken brain signals, which may lead to thinking and memory problems over time.

Introduction

Microplastics are tiny plastic particles that pollute the environment and have been found in human blood, lungs, and even the brain. The World Health Organization has identified the effects of microplastics on the brain as an important area for research. Understanding whether these particles harm brain cells is therefore important for human health.

One concern is that very small plastic particles are hydrophobic, meaning they repel water. This allows them to pass through the blood-brain barrier and dissolve into the fatty outer layer of neurons.[2] A neuron's outer layer acts like a small electrical capacitor, storing and releasing charge. This property is called membrane capacitance (C_m), measured in microfarads per square centimetre ($\mu\text{F}/\text{cm}^2$). If plastic particles make the membrane thicker, capacitance would rise, and the neuron would take longer to charge up before it can fire.

No study has yet directly modelled how plastic build-up affects the electrical behaviour of neurons. This project addresses that gap using the Hodgkin-Huxley model, first published in 1952, which mathematically describes how neurons produce electrical signals called action potentials.[1] By treating membrane capacitance as a variable, the study tests whether increasing

plastic load delays or weakens neural signals. The hypothesis is that higher plastic levels will slow down the first action potential and reduce its peak voltage.

Literature Review

The Hodgkin-Huxley model, published in 1952, is still the standard way to describe how neurons fire. Hodgkin and Huxley showed that action potentials come from the opening and closing of sodium (Na^+) and potassium (K^+) channels, controlled by three gating variables (m , h , and n). [1] Their equations accurately predicted the shape and speed of nerve signals in the squid giant axon, and the work earned them the 1963 Nobel Prize in Physiology or Medicine. The model is still used today because it can predict how changes to individual biological parameters affect neuron behaviour.

Recent research has started to look at how microplastics interact with cell membranes. Fleury and Baulin (2021) used molecular dynamics simulations to show that microplastic particles stretch and destabilise lipid membranes, changing their thickness and fluidity. [2] This supports the idea that embedded plastic particles would change the membrane's capacitance. Dayan and Abbott (2001) provide the standard computational method for solving Hodgkin-Huxley equations, showing that the forward Euler method works well when the time step is small enough. [3]

It was expected that increasing membrane capacitance would delay the start of action potentials and reduce their peak voltage. These predictions follow directly from the basic capacitor equation, in which charging time increases with capacitance for a fixed input current.

Method

Model Framework

The simulation models the electrical behaviour of a small section of a human neuron's membrane. The system is based on the Hodgkin-Huxley equations:

$$I = C_m \left(\frac{d_v}{d_t} \right) + \bar{g}_{Na} m^3 h (V - E_{Na}) + \bar{g}_k n^4 (V - E_k) + \bar{g}_L (V - E_L)$$

Here, I is the injected stimulus current in microamperes per square centimetre ($\mu\text{A}/\text{cm}^2$). V is the membrane potential in millivolts (mV). The variable t is time in milliseconds (ms). The variables m , h , and n are dimensionless gating probabilities that control how sodium (Na^+) and potassium (K^+) channels open and close. The maximum channel conductances were $\bar{g}_{Na} = 120.0$

millisiemens per square centimetre (mS/cm^2), $\bar{g}_K = 36.0 \text{ mS}/\text{cm}^2$, and $\bar{g}_L = 0.3 \text{ mS}/\text{cm}^2$. The equilibrium potentials were $E_{Na} = +50.0 \text{ mV}$, $E_K = -77.0 \text{ mV}$, and $E_L = -54.387 \text{ mV}$. The simulation started with $V = -65.0 \text{ mV}$, $m = 0.05$, $h = 0.6$, and $n = 0.32$.

Software Tools

All simulations were written in Python 3 using the NumPy library for calculations and Matplotlib for graphs. The code ran in a free Google Collaboratory notebook, so no special software was needed.

Simulation Procedure

Four scenarios were tested by changing the baseline membrane capacitance ($C_m = 1.0 \text{ }\mu\text{F}/\text{cm}^2$).

These were a healthy control neuron (1.0x) and three neurons with increasing plastic loads (1.5x, 2.0x, and 2.5x). This scaling acts as a mathematical stand-in for how plastic particles would thicken the membrane.

The equations were solved using the forward Euler method with a time step (Δt) of 0.01 ms over a total simulation time (T) of 60.0 ms. A $10.0 \text{ }\mu\text{A}/\text{cm}^2$ current was injected between 10.0 ms and 50.0 ms to make the neuron fire repeatedly. Each scenario ran the full simulation loop, updating every gating variable and recalculating the membrane voltage at each step.

Data Extraction

After each simulation, a peak-detection algorithm found every local maximum in the voltage trace above 0 mV. Three measurements were taken: the time of the first action potential (spike latency), the highest voltage reached (peak amplitude), and the total number of action potentials fired (spike count).

Results

Raw Data

The raw data are four voltage-versus-time arrays, one for each plastic load scenario. Each array contains 6,000 voltage measurements taken every 0.01 ms over the 60 ms simulation.

Extracted Data

The extracted data, produced by the peak-detection algorithm, are shown in Table 1.

Table 1: Electrophysiological measurements at different plastic load factors

Plastic Load Factor	Membrane Capacitance ($\mu F/cm^2$).	First Spike Latency (ms)	Peak Voltage Amplitude (mV)	Total Spike Count
1.0x (control)	1.0	12.11	40.73	3
1.5x	1.5	12.81	38.73	3
2.0x	2.0	13.50	36.49	3
2.5x (severe)	2.5	14.19	34.06	3

Data Visualisation

The voltage trace for the baseline neuron (1.0x plastic load) is shown in Figure 1. The neuron remains at its resting potential of -65 mV until the stimulus begins at 10 ms, then fires two clean action potentials before returning toward rest after the stimulus ends.

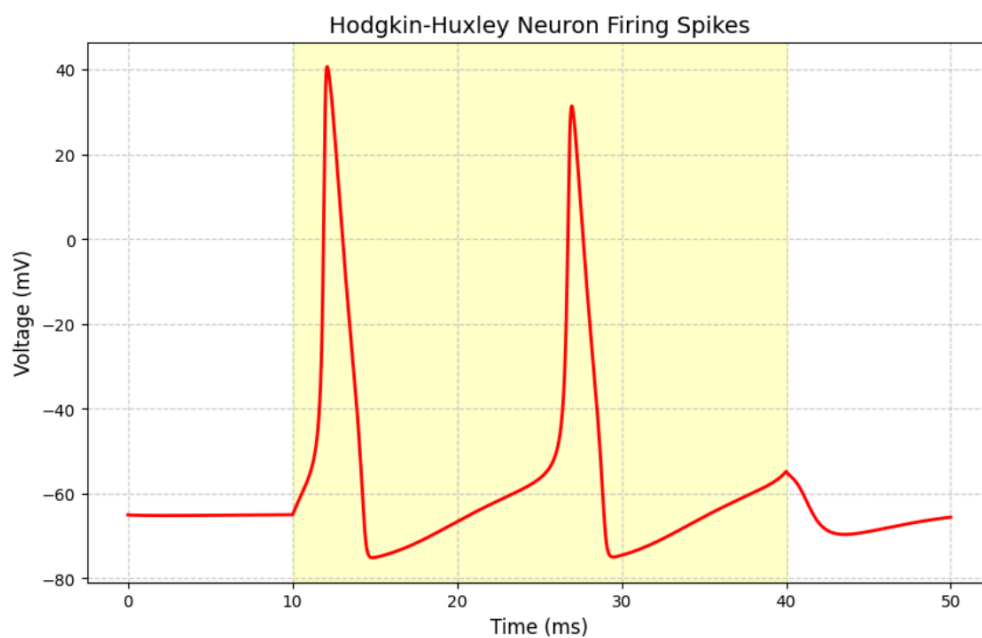


Figure 1: Baseline Hodgkin-Huxley neuron firing. Membrane voltage over a 50 ms simulation with a $10 \mu A/cm^2$ stimulus applied between 10 ms and 40 ms.

The voltage traces for all four plastic load scenarios are shown in Figure 2. The healthy control neuron (blue line) fires its first action potential at around 12 ms and reaches a peak of +40.73 mV. As the plastic load increases, the later traces (orange, green, and red) shift to the right and reach lower peaks.

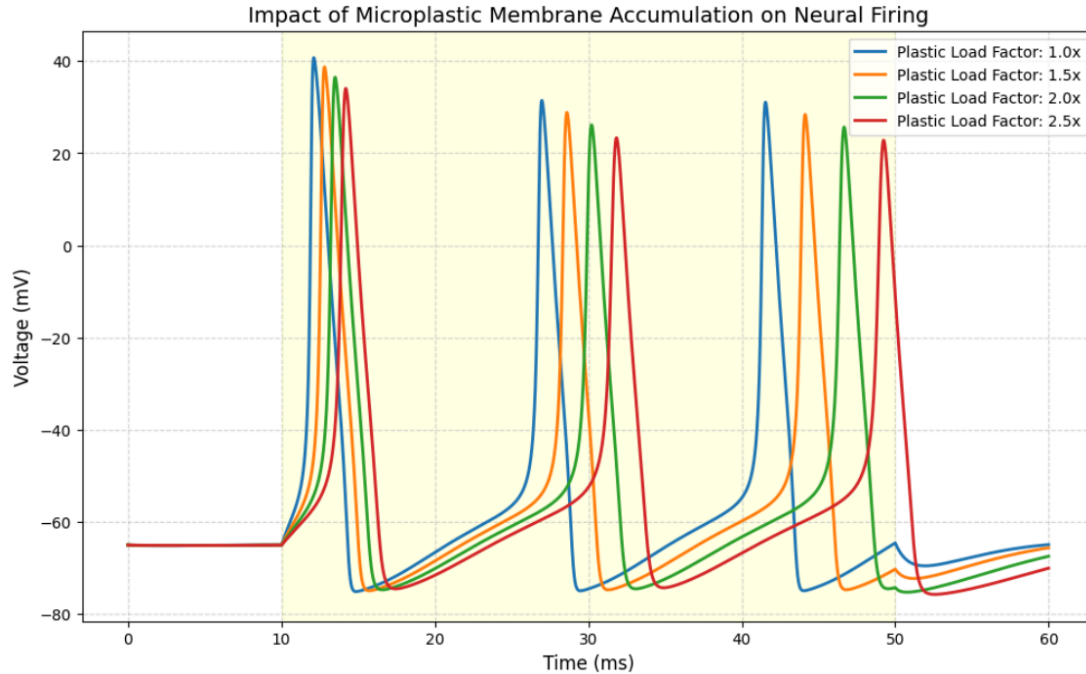


Figure 2: Effect of microplastic build-up on neuron firing. Voltage traces for four plastic load factors (1.0x, 1.5x, 2.0x, and 2.5x) over a 60 ms simulation. The shaded area shows the 10 $\mu\text{A}/\text{cm}^2$ stimulus window.

Analysis of Trends

Two clear trends appear in the data. First, spike latency increases with plastic load. The control neuron fires its first action potential at 12.11 ms, while the 2.5x neuron does not fire until 14.19 ms. This is a 17.18% delay. Second, peak voltage decreases with plastic load. The control neuron reaches 40.73 mV, while the 2.5x neuron only reaches 34.06 mV. This is a 6.67 mV drop, or a 16.38% reduction. The total spike count stayed at three for all four scenarios within the range tested.

Discussion / Analysis

The results support the hypothesis that increasing microplastic build-up reduces the quality of neural signals. Both latency and amplitude changed in a smooth, consistent way, matching what basic capacitor physics would predict.

Why the Signal Is Delayed

The 17.18% increase in first spike latency makes sense from the capacitor equation. For a fixed input current, the time needed to charge the membrane to threshold increases with capacitance. A 2.5x increase in C_m should theoretically cause a much bigger delay than 17.18%, but the neuron's gating kinetics partly compensate by adjusting their own speeds. Even so, a real neuron with this level of plastic contamination would take about 2 ms longer to send its first signal, which could add up over millions of neurons.

Why the Signal Is Weaker

The 6.67 mV drop in peak voltage reflects how much harder it is to charge a larger capacitor within a fixed time window. During the rising phase of the action potential, sodium channels must inject enough charge to overcome the membrane's capacitance. A thicker, more contaminated membrane holds more charge per unit voltage, leaving less driving force to push the voltage up. In a real neuron, a weaker action potential might fail to trigger neurotransmitter release at the synapse, effectively causing the signal to disappear.

What This Means for Medicine

The observed 17.18% delay in spike latency and 16.38% drop in peak amplitude resemble the mild signal problems seen in early nerve damage and low-level toxin exposure. If similar changes happen in living human neurons, they could slow down the senses, reduce the reliability of communication between neurons, and increase the energy cost of normal brain function. These effects would probably not cause obvious symptoms at low exposure levels, but build-up over years could contribute to memory and thinking problems or make existing brain diseases worse. The model therefore supports the need for more medical research into the effects of microplastics on the brain.

Comparison with Expected Outcomes

The results match what was predicted in the Literature Review. The mechanical stretching of membranes reported by Fleury and Baulin fits with the modelled rise in capacitance.[2] No published study has yet directly measured action potential latency in human neurons exposed to microplastics, so the direction and size of the effect reported here are a reasonable first estimate based on known biophysics.

Unusual Results

No unusual results appeared. All four scenarios fired three clean action potentials, and both measured values changed smoothly with the plastic load. This suggests the simulation is stable and the range tested does not cross any unexpected thresholds.

Sources of Error and Limits

Several limitations should be noted. First, the increase in membrane capacitance is a mathematical stand-in for plastic build-up, not a direct measurement, so the model is a prediction rather than a confirmation. Second, the simulation does not include biological responses such as myelin repair, glial activity, or changes in ion channel numbers, all of which would affect how a real neuron responds over time. Third, the Hodgkin-Huxley constants were originally measured in squid giant axons, so applying them to human neurons is an approximation. Fourth, the model assumes plastic particles are spread evenly across the membrane, whereas in reality their distribution would be uneven. Fifth, the forward Euler method is accurate for the small time step used but does introduce small numerical errors that could be reduced with a higher-order method.[3] Finally, the model represents a single small patch of membrane rather than the full branched structure of a real neuron, which would change the absolute values reported.

Possible Uses

The findings point to several practical uses. In medicine, this framework could help estimate exposure limits for brain harm, support the development of protective drugs, and inform public health guidelines on microplastic exposure. Environmental toxicology could use the same approach to assess brain risk in contaminated populations. Drug discovery teams could add plastic-load values when testing possible protective compounds.

Future Improvements

Future work should extend the simulation beyond 2.5x to find the exact plastic level at which the third action potential disappears. Adding temperature effects would make the model more realistic. Linking the simulation directly to molecular dynamics studies of plastic-lipid interactions would replace the current mathematical stand-in with physically measured capacitance values.[2] Ultimately, the predictions would need to be checked against real electrical recordings from human or animal tissue exposed to microplastics.

Conclusion

This project tested whether simulated microplastic build-up inside a neuron's membrane could weaken its electrical signals. Using a Hodgkin-Huxley model with scaled membrane capacitance, the most contaminated neuron (2.5x) fired its first action potential 17.18% later and reached a peak voltage 6.67 mV lower than the healthy control. These results support the hypothesis that plastic build-up reduces both the speed and strength of neural signals. The importance of the work lies in combining environmental toxicology with computational neuroscience, offering a first estimate of how microplastics could affect the brain. Limitations include the use of capacitance as a mathematical stand-in, the squid-derived origin of the Hodgkin-Huxley constants, and the absence of biological variability. The findings provide a starting point for future experimental and medical work, and they highlight the need to treat microplastic pollution as a potential long-term hazard to brain health.

Funding Statement / Acknowledgements

No external funding was received for this project. The author thanks their science teacher for guidance on computational methods and the developers of the open-source NumPy and Matplotlib libraries.

References

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